

Mini-Project NNs

https://git-disl.github.io/GTDLBench/datasets/mnist_datasets/

The number of categories of MNIST is 10, that is 0-9, 10 digits.

The Number of Samples per Category for MNIST

CATEGORY	0	1	2	3	4	5	6	7	8	9	TOTAL
#Training Samples	5,923	6,742	5,958	6,131	5,842	5,421	5,918	6,265	5,851	5,949	60,000
#Testing Samples	980	1,135	1,032	1,010	982	892	958	1,028	974	1,009	10,000

Samples



Mini-Project NNs

- ❑ Download the MNIST database and visualize.
- ❑ Randomly pick 10,000 training examples and 1,000 test
(verify the different digits are fairly represented)
(normalize the pixel values of the images to a range between 0 and 1 by dividing by 255)
- ❑ Build and train a Neural Network with two hidden layers of size 60 and 40, respectively.
- ❑ Evaluate the classification performance.
- ❑ Examine the receptive fields of neurons in the hidden layers.